We need **Maclaurin and Taylor polynomials** because they give **simple polynomial approximations** of complicated functions.

Maclaurin polynomial

Definition: If f can be differentiable n times at 0, then we define the nth Maclaurin polynomial for f to be

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

Example: Find the Maclaurin polynomials P_0, P_1, P_2, P_3 and P_n for e^x.

Sol^n:

let $f(x) = e^x$

$f'(x) = e^x$ ~~$= f''(x)$~~

$f''(x) = e^x = f'''(x) = \cdots = f^{(n)}(x)$

$f(0) = e^0 = 1$, $f'(0) = e^0 = 1 = f''(0) = f'''(0) = \cdots = f^{(n)}(0)$

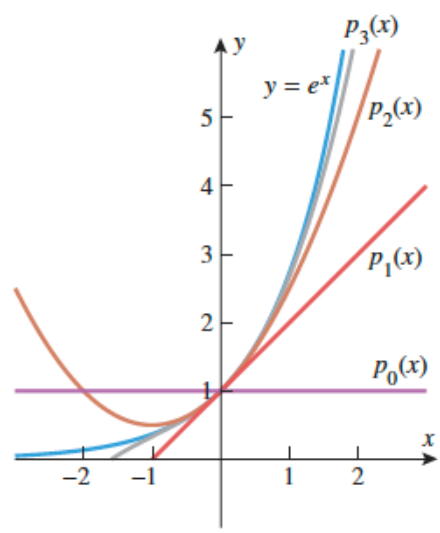
$P_0(x) = f(0) = 1$

$P_1(x) = f(0) + f'(0)x = 1 + 1.x = 1 + x$

$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 = 1 + x + \frac{1}{2!}x^2$

$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$

$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$



Therefore, $P_n(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}$

Taylor polynomial

Definition: If f can be differentiated n times at x_0, then we define the nth Taylor polynomial for f about $x = x_0$ to be

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

Example: Find the nth Taylor polynomial for $\frac{1}{x}$ about $x = 1$ and express it in sigma notation.

Solⁿ: Say, $f(x) = \frac{1}{x} = x^{-1}$ $\qquad f(1) = 1$

$f'(x) = -1 \cdot x^{-2}$ $\qquad f'(1) = (-1)\cdot 1 = -1$

$f''(x) = (-1)(-2)x^{-3}$ $\qquad f''(1) = (-1)(-2)\cdot 1 = 2!$

$= 2x^{-3}$

$f'''(x) = 2(-3)x^{-4}$ $\qquad f'''(1) = (-1)(-2)(-3)\cdot 1 = -3!$

$f^{iv}(x) = 2(-3)(-4)x^{-5}$ $\qquad f^{iv}(1) = (-1)(-2)(-3)(-4) = 4!$

$\vdots$

$f^{(n)}(x) = 2(-3)(-4)(-5)\cdots(-n)\, x^{-(n+1)}$, $\quad f^{(n)}(1) = (-1)(-2)\cdots(-n)\cdot 1$

$= (-1)^n n!$

So, we can write $f^{(k)}(x) = (-1)^k k!$

At $x = 1$, the Taylor polynomial for $1/x$:

$$\cancel{\sum_{k=0}^{n} (-1)^k (x-1)^k} =$$

$$\sum_{k=0}^{n} \frac{f^{(n)}(1)}{k!}(x-1)^k = \sum_{k=0}^{n} \frac{(-1)^k k!}{k!}(x-1)^k = \sum_{k=0}^{n} (-1)^k (x-1)^k$$

Extra Problems:

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7–16 Find the Maclaurin polynomials of orders $n = 0, 1, 2, 3,$ and 4, and then find the nth Maclaurin polynomials for the function in sigma notation.

7. e^{-x} **8.** e^{ax} **9.** $\cos \pi x$

10. $\sin \pi x$ **11.** $\ln(1 + x)$ **12.** $\dfrac{1}{1+x}$

13. $\cosh x$ **14.** $\sinh x$ **15.** $x \sin x$

16. xe^x

17–24 Find the Taylor polynomials of orders $n = 0, 1, 2, 3,$ and 4 about $x = x_0$, and then find the nth Taylor polynomial for the function in sigma notation.

17. $e^x;\ x_0 = 1$ **18.** $e^{-x};\ x_0 = \ln 2$

19. $\dfrac{1}{x};\ x_0 = -1$ **20.** $\dfrac{1}{x+2};\ x_0 = 4$

21. $\sin \pi x;\ x_0 = \dfrac{1}{2}$ **22.** $\cos x;\ x_0 = \dfrac{\pi}{2}$

23. $\ln x;\ x_0 = 1$ **24.** $\ln x;\ x_0 = e$